

Wenlong Ding (Curriculum Vitae)

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EDUCATION

The Chinese University of Hong Kong (CUHK)

Shatin, Hong Kong SAR, China

Ph.D. in Computer Science and Engineering

08/2021 – Present

- Thesis (Ongoing): Learning-Inspired Network Traffic and Configuration Management.

Huazhong University of Science and Technology (HUST)

Wuhan, Hubei, China

B.E. in Computer Science and Technology (ACM class)

09/2017 – 06/2021

- GPA: 90.64/100 or 3.96/4.0 | Rank: Top 5%

RESEARCH INTERESTS

Topic 1: Learning/LLM-Driven Network Configuration Management

05/2023 – Present

- **Agentic Routing Path Analysis (RPA) at Scale:** Benchmark [C1]; agentic RPA system for global production datacenter networks [A2*].
- **Intent-Based Configuration:** ACL configuration generation and deployment [A1]; configuration template database [C3].

Topic 2: Learning-Based Traffic Engineering (TE) Systems

08/2021 – 04/2023

- Link failure restoration in TE [C2]; TE acceleration [J1].

PUBLICATION

Conference Papers

[C1] **Wenlong Ding**, Zhixiong Niu, Jianan Yang, Fajun Zhang, Bo Zhang, Ling Liang, Yongqiang Xiong, Tianyin Xu, and Hong Xu. RoutingBench: Can Agentic Routing Analysis Scale to Production Datacenter Networks? In *Advances in Neural Information Processing Systems (NeurIPS), Evaluations and Datasets Track*, 2026 (To appear).

[C2] **Wenlong Ding** and Hong Xu. Dynamic Learning-based Link Restoration in Traffic Engineering with Archie. In *Proceedings of the IEEE International Conference on Computer Communications (IEEE INFOCOM)*, 2024.

[C3] **Wenlong Ding**, Libin Liu, Li Chen, and Hong Xu. ABC: Automatic Bottom-up Construction of Configuration Knowledge Base for Multi-Vendor Networks. In *Proceedings of the Fifth IEEE International Conference on Cognitive Machine Intelligence (IEEE CogMI) (Invited Paper)*, 2023.

[C4] Jianqiang Li, **Wenlong Ding**, Zhixiong Niu, Yongqiang Xiong, and Hong Xu. Saving Lost Intents in Network Configurations. In *Proceedings of the 10th Asia-Pacific Workshop on Networking (ACM APNet)*, 2026.

Journal Papers

[J1] **Wenlong Ding**, Libin Liu, Li Chen, and Hong Xu. Learning for Accelerated Traffic Engineering with Differentiable Network Modeling. In *IEEE Transactions on Network Science and Engineering (TNSE)*, vol. 13, pp. 25–36, 2026.

Posters

[P1] **Wenlong Ding**, Jianqiang Li, Zhixiong Niu, Huangxun Chen, and Hong Xu. Poster: Automating Network Configuration with Natural Language Intents. In *Proceedings of the ACM SIGCOMM 2024 Conference: Posters and Demos (ACM SIGCOMM Posters and Demos)*, 2024.

Preprints and Manuscripts Under Review

[A1] (Under review) **Wenlong Ding**, Jianqiang Li, Zhixiong Niu, Huangxun Chen, Yongqiang Xiong, and Hong Xu. Automating Conflict-Aware ACL Configurations with Natural Language Intents. *arXiv preprint arXiv:2508.17990*, 2025.

[A2*] (Under review) TARS: Scaling Agentic Routing Analysis for Production Datacenter Networks.

- Scaling agent skills for *RPA* across global production datacenter networks through context management and symbolic reflection, guided by operator usability requirements and operational experience.

[A3*] (Under review; *non-first author*) Adaptive Route Scaling for Cloud WANs.

- Reducing FIB entries while maintaining effective load balancing across the global cloud WAN.

* *These projects were conducted by **Wenlong Ding** during his internship at Microsoft Research Asia (MSRA), in collaboration with Azure network operators.*

INTERNSHIP

Research Intern — Microsoft Research Asia, Beijing, China

02/2025 – 07/2026

- *Lead role*: Resolved RPA scalability challenges in global datacenter networks to support Azure network operators' daily analysis and diagnosis [C1, A2*].
- *Contributing role*: Assisted with writing a paper for the project on FIB optimization to reduce memory pressure in Azure's global WAN. [A3*]

SERVICE AND TEACHING

Academic Services

Reviewer	IEEE Transactions on Network Science and Engineering.
Shadow TPC	ACM CoNEXT 2025 2026.
Artifact Evaluation Committee	USENIX NSDI 2026 2027, ACM SIGCOMM 2026, ACM EuroSys 2026, ACM ASPLOS 2026.

Teaching Assistant at CUHK

CSCI4430	<i>Data Communication and Computer Networks</i> . Spring 2023; Fall 2023.
ENGG2020	<i>Digital Logic and Systems</i> . Fall 2022; Spring 2024.
CSCI3170	<i>Introduction to Database System</i> . Fall 2024.
ENGG1103	<i>Multivariable Calculus for Engineers</i> . Spring 2022.
ENGG2760	<i>Probability for Engineers</i> . Fall 2021.

HONORS AND AWARDS

06/2021	Excellent Undergraduate Thesis (3%), HUST
05/2021	Outstanding Graduate and Honours Bachelor's Degree (3%), HUST
09/2020	Merit Student (5%), HUST
12/2019	Outstanding Individual in University Students Innovation and Entrepreneurship Project, HUST
04/2019	Meritorious Winner, Mathematical Contest in Modeling (5%), COMAP, USA